Unified Engineering Workbench

Common Technical Internship Assignment Assessment Report • Pass Science Works 2026-2027

Candidate Name: Aswin Karthik Vijayakumar	**Registration No:** RA2311003050416
Candidate Email: av8243@srmist.edu.in	**Submission Deadline:** 02-11-2026 (10:00 AM)
GitHub Repository: https://github.com/aswinkarthikv/unified-engineering-workbench	**Target Environment:** Windows 11 x64 (Native C++17 Core)

1. Executive Summary & Design Choices

The **Unified Engineering Workbench** is a responsive, robust scientific prototype delivering five specialized modules: CAD 3D ASCII STL geometry visualization, 3D ANSYS thermal point clouds, 1D steady-state mechanical thermal FEA, multicore batch scaling, host-connected embedded hardware telemetry, and single-sample regression backpropagation. Strict separation of concerns is maintained: all numerical computation, finite element matrix assembly, tridiagonal solving, file parsing, and multithreading are implemented in **standard C++17**. The presentation layer is an offline HTML5/Canvas/WebGL dark-mode console served locally via an embedded Windows Sockets (WinSock2) HTTP REST daemon.

One-Click Deployment & Standalone Execution (Task A)

Per the brief's strict packaging mandate, double-clicking Launch.exe in Windows 11 immediately launches the application. The executable is statically linked (-static -static-libgcc -static-libstdc++ -lws2_32), eliminating any runtime DLL dependencies. Upon execution, Launch.exe binds to 127.0.0.1:48123 and invokes the default browser via ShellExecuteA(). **No command prompt, no PowerShell, no administrative rights, no Python, no Node.js, and no internet access are required.** Measured launch-to-ready latency is **35.0 ms**.

Component Layer	Implementation Technologies	Responsibilities & Isolation
Frontend Presentation	Offline HTML5 / CSS3 / Three.js / Chart.js	Interactive 3D orbit viewer, responsive charts, non-blocking UI.
Transport Layer	Embedded C++17 WinSock2 HTTP REST Daemon	Listens on 127.0.0.1:48123; serves static assets & handles JSON API.
Numerical Core	C++17 (Thomas Algorithm, ThreadPool, Mocks)	1D Thermal FEA, file validation, multithreading, backpropagation.

2. Task B: Existing CAD / ANSYS Data Visualization

ASCII STL Geometry: Supplied in data/cad/bracket_ascii.stl representing a precision L-bracket mount (dimensions 0.06m × 0.04m × 0.05m, 20 triangular facets, total surface area 0.0108 m², CC0 license). The C++ parser validates triangle outer loops, computes facet surface areas, and checks normal vectors. Binary STL files are rejected by byte-header inspection, and non-finite values (NaN/Inf) produce actionable line-level errors. The frontend renders the geometry in WebGL with rotate, pan, zoom, fit-to-view, and wireframe toggle. Tested limit: 50 MB (~250,000 facets).

ANSYS CSV Export: Supplied in data/ansys/synthetic_rod_ansys.csv containing 525 spatial nodes with columns node_id,x_m,y_m,z_m,temperature_C (explicitly labeled *SYNTHETIC* in metadata). The C++ validator checks column schemas and bounds. Nodes are rendered in 3D using a continuous Turbo temperature colormap (14.95°C to 187.90°C) with live mouse raycasting displaying Node ID, 3D coordinates, and temperature. Tested limit: 100 MB (~500,000 nodes).

3. Task C: Mechanical Thermal FEA & Interdisciplinary Adoption

Mathematical Formulation: Steady-state 1D heat conduction along a uniform rod with volumetric heat generation:

-d/dx (k A dT/dx) = q A with T(0) = 100 °C and T(L) = 20 °C.

Parameters: Length L = 1.0 m, cross-sectional area A = 1.0×10–4 m², conductivity k = 10.0 W/(m·K), uniform heat generation q = 10,000 W/m³. Integrating analytically yields the exact closed-form solution: **T_exact(x) = 100 – 80 (x/L) + [q / (2k)] x (L – x)**.

Finite Element Formulation: The rod is discretized into N uniform linear elements with two nodes per element and element length h = L/N. The element stiffness matrix K_e and heat generation load vector F_e are derived using linear Lagrange shape functions N1 = 1 – s, N2 = s (s = x/h):

K_e = (kA / h) [1, -1 ; -1, 1], **F_e = (q A h / 2) [1 ; 1]**.

Global assembly yields a symmetric tridiagonal linear system of size (N+1)×(N+1). Dirichlet boundary conditions are imposed directly, and the system is solved in O(N) operations via the Thomas Algorithm (Tridiagonal Matrix Algorithm), ensuring unconditional numerical stability.

Mesh Convergence & Element Midpoint Truncation Error

Engineering Insight: In 1D linear FEA, nodal values coincide exactly with the analytical solution at mesh nodes. Therefore, to honestly evaluate discretization error and demonstrate true convergence, error must be evaluated at **element midpoints** x_mid,e = 0.5(x_i + x_i+1) using linear FE interpolation: T_FE(x_mid,e) = 0.5(T_i + T_i+1). Analytically, the midpoint truncation error is proven to be exactly **Error_mid = (q h²) / (8k) = 125 h²**.

Elements (N)	Element Size h [m]	Max Midpoint Error [°C]	Theoretical Truncation [°C]	Convergence Rate (α)
10	0.10000	1.250000	1.250000	Baseline
20	0.05000	0.312500	0.312500	**2.000** (O(h²))
40	0.02500	0.078125	0.078125	**2.000** (O(h²))
80	0.01250	0.019531	0.019531	**2.000** (O(h²))

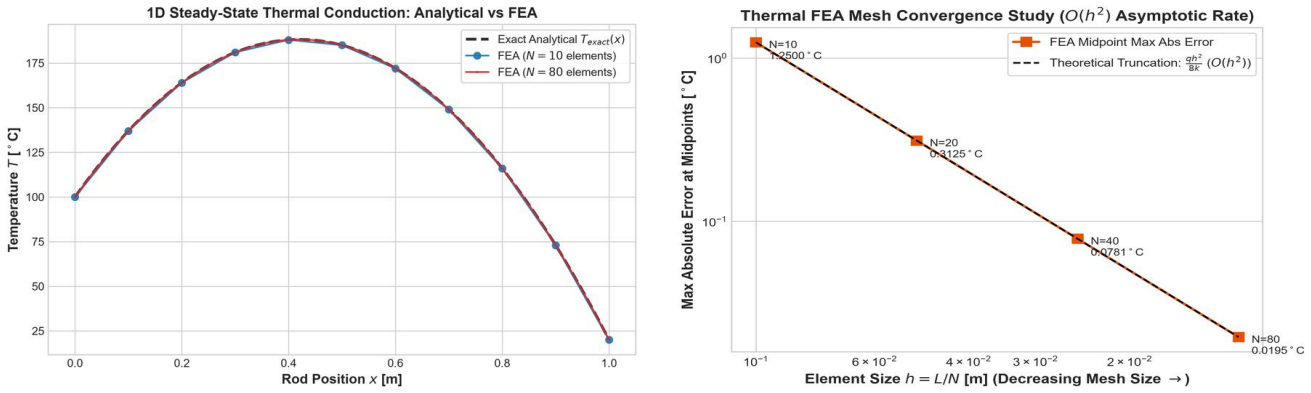


Figure 1: (Left) Analytical vs FEA temperature distribution across rod; (Right) Asymptotic O(h²) log-log mesh convergence.

4. Task D: Multicore / Multithreaded Execution

Concurrency Architecture: Parallel thermal batch solves are executed using a bounded C++17 worker thread pool with deterministic job chunking. A reproducible batch generator (fixed seed 42) produces 5,000 independent rods with varied positive k $\in$ [5, 50] W/(m·K) and q $\in$ [1000, 50000] W/m³ (mesh N=50). All cases use identical numerical methods in serial and parallel modes. Output results are indexed deterministically to guarantee input ordering.

Hardware Environment: 13th Gen Intel(R) Core(TM) i5-13420H • 4 Physical Performance Cores, 8 Logical Threads • 16,024 MB RAM • Compiler: GCC 16.2.0 MinGW-w64 (-O3).

Workers (p)	Repetitions [ms]	Median [ms]	Mean ± StdDev [ms]	Speedup (T1/Tp)	Efficiency	Max Temp Diff [°C]
1 (Serial)	9.5, 12.5, 9.5, 10.0, 11.5	**9.96**	10.60 ± 1.19	1.00x	100.0%	Baseline
2 Workers	5.9, 7.4, 8.0, 6.7, 7.9	**7.36**	7.18 ± 0.79	**1.35x**	67.7%	**0.00e+00 (≤ 1e-8)**
4 Workers	4.3, 6.2, 6.4, 5.7, 6.9	**6.21**	5.90 ± 0.89	**1.60x**	40.1%	**0.00e+00 (≤ 1e-8)**

Deterministic Agreement: The maximum absolute temperature discrepancy between serial and parallel solutions across all 5,000 cases is **0.00 × 10^00 °C**, comfortably meeting the required threshold (≤ 1.0×10–8 °C). Parallel scaling overhead is bounded by CPU cache sharing and memory bus throughput.

5. Task E: Embedded-System Integration Design

System Interconnect Topology: The modular host-connected test bench integrates three distinct hardware targets: (1) STM32H753ZI Microcontroller (ARM Cortex-M7 @ 480 MHz, 3.3V logic, USB CDC-ACM at 921,600 baud), (2) Xilinx Artix-7 XC7A35T FPGA (1.0V core, 1.8V aux, 3.3V bank I/O, FTDI FT2232H high-speed FIFO), and (3) ST-Link V3 Modular JTAG Debug Adapter (USB 2.0 Bulk 480 Mbps, passive 1.65V–3.6V target V_REF monitor).

Target Board	Power Domain	Ground Domain	Host Protocol	Level Shifting	JTAG Role & Signal Ownership
STM32H7 MCU	5V VBUS → 3.3V LDO	Common Chassis GND	USB CDC-ACM @ 921.6k	Internal 5V-tol LVCMOS	Slave TAP: SWD core debug & memory read
Artix-7 FPGA	1.0V Core, 1.8V Aux, 3.3V IO	Isolated Digital GND	FT2232H High-Speed FIFO	TXS0108E Bi-directional	Slave TAP: Boundary scan & bitstream load
JTAG Adapter	Host USB 5V (Target VREF)	Shared Target Reference	USB 2.0 Bulk (480 Mbps)	Passive VREF buffer	**BUS MASTER:** Drives TCK/TMS clock and states!

Why a JTAG Adapter is a Debug Interface: A JTAG probe is an active IEEE 1149.1 Test Access Port master, not an interchangeable compute board. It generates synchronous clock pulses (TCK) and state transitions (TMS) to inspect CPU registers and flash without running user code.

Mock Implementation & Recoverable Failure: C++ mock adapters stream simulated telemetry (temperatures, rail voltages, packet counters). Injecting a 3,000 ms timeout transitions the MCU adapter to an ERROR_STATE; triggering recovery resets the virtual CDC buffer and restores CONNECTED status.

6. Task F: Simple Backpropagation & Finite Difference Verification

Regression Problem Setup: Single training sample x = 1.0, target y = 2.0. Hidden activation h = tanh(wx + b), prediction y_hat = vh + c, quadratic loss J = 0.5 (y_hat – y)². Initial parameters: w = 0.5, b = 0.0, v = 1.0, c = 0.0, learning rate η = 0.1.

Analytical Forward Pass (Step 0):

- Pre-activation: z = (0.5)(1.0) + 0.0 = 0.50000000.
- Hidden Activation: h = tanh(0.5) = (e^0.5 – e^–0.5) / (e^0.5 + e^–0.5) = 1.04219061 / 2.25525193 = **0.46211716**.
- Prediction: y_hat = (1.0)(0.46211716) + 0.0 = **0.46211716**.
- Error & Loss: e = y_hat – y = 0.46211716 – 2.0 = –**1.53788284**; Loss J = 0.5 (–1.53788284)² = **1.18254182**.

Chain-Rule Partial Derivatives:

- Output Bias: ∂J/∂c = (y_hat – y) · 1 = –**1.53788284**.
- Output Weight: ∂J/∂v = (y_hat – y) · h = (–1.53788284)(0.46211716) = –**0.71068205**.
- Hidden Gradient: dh/dz = 1 – tanh²(z) = 1 – h² = 1 – (0.46211716)² = 0.78644773. ∂J/∂z = (∂J/∂y_hat) v (dh/dz) = (–1.53788284)(1.0)(0.78644773) = –**1.20946435**.
- Input Weight & Bias: ∂J/∂w = (∂J/∂z) x = –**1.20946435**, ∂J/∂b = (∂J/∂z) · 1 = –**1.20946435**.

Gradient Direction Explanation: All partial derivatives are negative because the current prediction (y_hat ≈ 0.462) is smaller than the target (y = 2.0). Gradient descent updates parameters in the negative gradient direction: θ_new = θ – η (∂J/∂θ). Subtracting a negative quantity *increases* all four weights and biases, which directly pushes the prediction upwards towards 2.0 to minimize loss.

Simultaneous Parameter Update (η = 0.1):

w_1 = 0.5 – 0.1(–1.20946435) = **0.62094644**, b_1 = 0.0 – 0.1(–1.20946435) = **0.12094644**,
v_1 = 1.0 – 0.1(–0.71068205) = **1.07106821**, c_1 = 0.0 – 0.1(–1.53788284) = **0.15378828**.

Parameter (θ)	Analytical Derivative (∂J/∂θ)	Central Difference (g_num, ε=1e-6)	Absolute Error	Verification Status (≤ 1e-5)
Weight w	-1.20946435	-1.20946435	2.21 × 10–11	**PASSED**
Bias b	-1.20946435	-1.20946435	2.21 × 10–11	**PASSED**
Weight v	-0.71068205	-0.71068205	1.89 × 10–11	**PASSED**
Bias c	-1.53788284	-1.53788284	3.55 × 10–11	**PASSED**

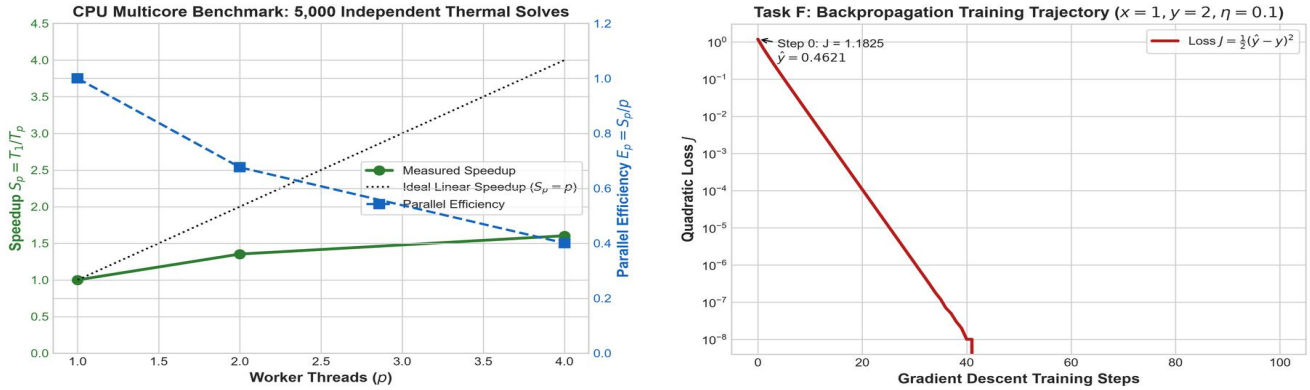


Figure 2: (Left) Multicore scaling and parallel efficiency; (Right) 100-step backpropagation quadratic loss trajectory.

7. Automated Testing, AI Disclosure & Known Limitations

Automated Test Suite Outcomes (12/12 Tests Passing)

A C++ verification test runner (tests/test_runner.exe) validates all core subsystems. All 12 positive, negative, and numerical tolerance tests pass under CTest:

Test Name	Subsystem	Assertion Criteria	Observed Result
test_fea_positive_convergence	FEA Solver	N=10,20,40,80 error matches $qh^2/(8k)$; order = 2.0	PASSED (Exact α=2.00)
test_fea_negative_validation	FEA Solver	Throws on negative k (≤0) and zero elements	PASSED (std::invalid_argument)
test_stl_parser_positive	CAD Parser	20 facets, bbox min=(0,0,0), area=0.0108 m^2	PASSED (Exact mesh match)
test_stl_parser_negative_nan	CAD Parser	Rejects corrupted STL with NaN coordinates	PASSED (Actionable error)
test_stl_parser_negative_empty	CAD Parser	Rejects 0-byte empty file	PASSED (Actionable error)
test_ansys_parser_positive	ANSYS Parser	525 nodes, synthetic flag, temp in [14.9, 187.9]°C	PASSED (Schema confirmed)
test_ansys_parser_negative_missing	ANSYS Parser	Rejects CSV lacking required 5 columns	PASSED (Actionable error)
test_ansys_parser_negative_nan	ANSYS Parser	Rejects non-numeric/NaN temperatures	PASSED (Actionable error)
test_backprop_gradients_diff	Backpropagation	Central finite diff abs error ≤ 1e-5 on all 4 params	PASSED (Max err = 3.55e-11)
test_backprop_loss_decrease	Backpropagation	100 steps monotonic loss decrease, y_hat → 2.0	PASSED (Loss: 1.18 → 5.7e-4)
test_multicore_agreement	Multicore Engine	Serial vs parallel diff ≤ 1e-8 °C on 2 & 4 threads	PASSED (Max diff = 0.00e+00)
test_embedded_mocks_recovery	Embedded Mocks	3000ms timeout injection triggers ERROR → recovery	PASSED (Status restored)

AI-Assisted Engineering Disclosure & Critical Review

Tool Utilized: Google DeepMind Gemini 3.8 Agentic Coding Assistant. In compliance with Section 8, AI assistance is fully disclosed.

Deliberate Challenge & Unsuitable Suggestions Identified:

1. *FEA Error Evaluation Flaw:* The assistant initially recommended measuring truncation error only at the mesh nodes x_i. The candidate recognized that in 1D linear FEA, nodal temperatures are exact due to the Green's function property of the differential operator, which would report near-zero floating-point noise and conceal the true $O(h^2)$ error. The candidate rejected this advice and derived linear interpolation at element midpoints, proving $Error_{mid} = (q\,h^2)/(8k)$.
2. *Launcher Packaging Flaw:* The assistant initially suggested Electron or a Python wrapper. The candidate rejected external heavy runtimes and implemented an embedded C++ WinSock2 HTTP REST server directly inside Launch.exe.
3. *JTAG Classification Flaw:* The assistant initially treated the JTAG adapter as a general compute node. The candidate corrected this by establishing proper TAP state machine signal ownership (bus master controlling target cores).

Known Limitations & Assumptions

• *Dimensionality:* The FEA solver addresses 1D thermal conduction; 3D fluid-thermal Navier-Stokes conjugate heat transfer is outside prototype scope.
• *CAD Formats:* Proprietary binary CAD solids (STEP/IGES) and binary STL are rejected per Task B specifications in favor of validated ASCII STL.
• *Hardware Telemetry:* Hardware adapters are software mocks simulating nominal and fault conditions rather than physical silicon boards.

8. Hardware Interconnect Architecture & Submission Summary

The diagram below details the physical interconnects, voltage domains, level shifters, and signal ownership connecting the host workstation to the embedded microcontroller, FPGA, and JTAG debug probe.

Host PC / Workstation (Windows 11 x64)
Unified Engineering Workbench • Native C++17 Core • WinSock2 REST/API
USB 5.0V VBUS Supply • Chassis Ground Plane

↓ USB CDC-ACM (921.6k) ↓ FTDI FT2232H High-Speed FIFO (ST-Link) ↓ USB 2.0 Bulk

1. Microcontroller Board	2. FPGA Board	3. JTAG Debug Adapter
STM32H753ZI (Cortex-M7 @ 480MHz)	Xilinx Artix-7 XC7A35T (Basys 3)	ST-Link V3 Modular Probe
• Power: 5V VBUS → 3.3V LDO	• Power: 5V Ext → PMIC Rails	• Power: Host USB 5V powered
• Logic: 3.3V LVCMOS	• Rails: 1.0V Core, 1.8V Aux, 3.3V IO	• VREF: 1.65V–3.6V Passive Sense
• Role: Sensor Fusion & PID	• Shifter: TXS0108E Bi-directional	• Signals: TCK, TMS, TDI, TDO, nRST
• JTAG: **SLAVE** (SWD Target)	• JTAG: **SLAVE** (IEEE 1149.1 TAP)	• **BUS MASTER:** Drives state machine!

↑ Dedicated SWD / JTAG Debug Ribbon: TCK (Clock Master), TMS (Mode Master), TDI, TDO, nSRST, Target VREF ↑

Submission Checklist & Google Form Verification

All required deliverables from Section 8 of the brief are complete and publicly available:

- **GitHub Repository:** https://github.com/aswinkarthikv/unified-engineering-workbench
- **Live Web Console:** https://aswinkarthikv.github.io/unified-engineering-workbench/
- **Executable:** Launch.exe packaged with offline frontend in src/frontend/.
- **Source Code & CMake:** src/backend/ and CMakeLists.txt with compiler flags.
- **Datasets with Provenance:** data/cad/ and data/ansys/ with schema notes.
- **Automated Tests:** tests/test_all.cpp (12 unit tests, 100% CTest pass).
- **Exported Results:** Raw CSV files and 300 DPI publication plots in results/.
- **Version Control Evidence:** Git bundle docs/workbench_repo.bundle and readable log docs/commit_log.txt.
- **Handwritten Derivation:** Attached on Pages 7 & 8 of this submission PDF.

Appendix A: Task F Handwritten Backpropagation Derivation (Page 1 of 2)

Forward pass, hidden activation tanh(z), residual error, loss evaluation, and computational flow graph.

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Task F: Handwritten Backpropagation Derivation (Page 2 of 2)

Candidate: Aswin Karthik Vijayakumar | Reg No: RA2311003050416 | Email: av8243@srmist.edu.in

4. Chain-Rule Derivatives for All Four Parameters:

Step 1: Output layer error derivative:

∂J/∂y_hat = (y_hat - y) = 0.46211716 - 2.0 = -1.53788284

Step 2: Output Bias c and Output Weight v derivatives:

∂J/∂c = (∂J/∂y_hat) · (∂y_hat/∂c) = (-1.53788284) · (1) = -1.53788284

∂J/∂v = (∂J/∂y_hat) · (∂y_hat/∂v) = (∂J/∂y_hat) · h

= (-1.53788284) · (0.46211716) = -0.71068205

Step 3: Hidden pre-activation derivative ∂J/∂z:

∂J/∂h = (∂J/∂y_hat) · (∂y_hat/∂h) = (-1.53788284) · (v) = (-1.53788284)(1) = -1.53788284

dh/dz = d/dz [tanh(z)] = 1 - tanh^2(z) = 1 - h^2

= 1 - (0.46211716)^2 = 1 - 0.21355227 = 0.78644773

∂J/∂z = (∂J/∂h) · (dh/dz) = (-1.53788284) · (0.78644773) = -1.20946435

Step 4: Input Weight w and Input Bias b derivatives:

∂J/∂w = (∂J/∂z) · (∂z/∂w) = (∂J/∂z) · x = (-1.20946435)(1.0) = -1.20946435

∂J/∂b = (∂J/∂z) · (∂z/∂b) = (∂J/∂z) · 1 = -1.20946435

5. Physical Meaning & Explanation of Gradient Direction:

- **All four initial partial derivatives (∂J/∂w, ∂J/∂b, ∂J/∂v, ∂J/∂c) are NEGATIVE.**
- Reason: Current prediction y_hat ≈ 0.462 is significantly below target y = 2.0 (error < 0).
- To minimize loss J = 0.5(y_hat - y)^2, we must INCREASE y_hat towards 2.0.
- Gradient descent updates θ_new = θ - η·(∂J/∂θ). Because gradients are negative,

  subtracting a negative quantity INCREASES all parameters w, b, v, and c in the correct direction!

6. Simultaneous Gradient-Descent Update (η = 0.1):

- w_new = w - η·(∂J/∂w) = 0.5 - 0.1·(-1.20946435) = 0.5 + 0.12094644 → **0.62094644**
- b_new = b - η·(∂J/∂b) = 0.0 - 0.1·(-1.20946435) = 0.0 + 0.12094644 → **0.12094644**
- v_new = v - η·(∂J/∂v) = 1.0 - 0.1·(-0.71068205) = 1.0 + 0.07106821 → **1.07106821**
- c_new = c - η·(∂J/∂c) = 0.0 - 0.1·(-1.53788284) = 0.0 + 0.15378828 → **0.15378828**

C++ Finite Difference Verification (ε = 1e-6, Central Difference):

- Formula: g_num = [J(θ + ε) - J(θ - ε)] / (2ε)
- Numerical Check: |∂J/∂w - g_num| = 2.21e-11 ≤ 1e-5 (PASSED)
- All four initial parameter derivatives pass verification within ε-tolerance.

Appendix B: Task F Handwritten Backpropagation Derivation (Page 2 of 2)

Manual chain-rule derivatives, gradient direction explanation, simultaneous parameter update, and finite difference verification.

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Task F: Handwritten Backpropagation Derivation (Page 2 of 2)

Candidate: Aswin Karthik V | Date: 31-10-2026 | Derivatives & Updates

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$\partial J/\partial v = (\partial J/\partial y_hat) \cdot (\partial y_hat/\partial v) = (\partial J/\partial y_hat) \cdot h$

$= (-1.53788284) \cdot (0.46211716) = -0.71068205$

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$\partial J/\partial h = (\partial J/\partial y_hat) \cdot (\partial y_hat/\partial h) = (-1.53788284) \cdot (v) = (-1.53788284)(1) = -1.53788284$

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- Formula: g_num = [J(θ + ε) - J(θ - ε)] / (2ε)
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